METRIC

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DETAIL SPECIFICATION

STRONTIUM NITRATE, ANHYDROUS

Reactivated after 10 October 2023 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for two grades of anhydrous strontium nitrate and is intended for use in ammunition (see 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

Comments, suggestions, or questions on this document should be addressed to: US Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or email to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 6810

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MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY

Reagent Chemicals	-	Specifications and Procedures for Reagents and Standard Grade Reference Materials

(Copies of this document are available from https://pubs.acs.org/.)

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Wire Cloth and Sieves for Testing Purposes
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM D1193	-	Reagent Water

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise specified in this document or in the contract, in the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Strontium nitrate assay (Grades A and B). Strontium nitrate, including any barium nitrate, shall have an assay of not less than 99.5 weight percent (wt.%).

3.3 Moisture (Grades A and B). Moisture content of strontium nitrate shall be not greater than 0.05 weight percent.

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3.4 Hygroscopicity (Grades A and B). Hygroscopicity of strontium nitrate shall be not greater than 0.05 weight percent.

3.5 Acidity (pH) (Grades A and B). Acidity of strontium nitrate shall be not less than 6.0 and not greater than 7.0.

3.6 Water-insoluble matter (Grades A and B). Water-insoluble matter of strontium nitrate shall be not greater than 0.02 weight percent.

3.7 Grit (Grades A and B). Grit content of strontium nitrate shall be not greater than 0.02 weight percent.

3.8 Chloride (Grades A and B). Chloride content (as Cl) of strontium nitrate shall be not greater than 0.003 weight percent. When alternate test method is used to verify compliance the sample shall be not more opaque than a solution prepared to this concentration.

3.9 Ammonium compounds (Grades A and B). Ammonium compounds (as NH_3) of strontium nitrate shall be not greater than 0.01 weight percent.

3.10 Chemical impurities. The strontium nitrate shall comply with the chemical impurities requirements specified in Table I.

TABLE I. Chemical impurities requirements.

Property	Requirement	
	Grade A	Grade B
Barium, wt. %, maximum (max.) ……..	0.05	1.5
Calcium, wt. %, max. …….	0.003	0.003
Magnesium, wt. %, max. ……...	0.003	0.003
Sodium, wt. %, max. ……..	0.003	0.003
Iron, wt. %, max. ……...	0.003	0.003
Bismuth, wt. %, max. …….	0.003	0.003
Lead, wt. %, max. ……..	0.003	0.003
Copper, wt. %, max. ……..	0.003	0.003
Sulfur (total), wt. %, max. …….	0.01	0.01

3.11 Granulation.

3.11.1 Grade A. The granulation requirements for Grade A strontium nitrate shall be not less than 95.0 weight percent through a number (No.) 100 U.S. Alternative sieve when tested in accordance with 4.4.11. The sieve used shall conform to ASTM E11.

3.11.2 Grade B. The granulation requirements for Grade B strontium nitrate shall be not less than 99.9 weight percent through a No. 50 U.S. Alternative sieve, and not greater than 30.0 percent through a No. 140 sieve. The sieves used shall conform to ASTM E11.

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3.12 Workmanship. The strontium nitrate shall be uniform in appearance and quality and free from dust, debris, visible impurities, or foreign matter.

4. VERIFICATION

TABLE II. Requirement/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Strontium nitrate assay	X		X	X	X	X	4.4.1
3.3	Moisture	X		X	X	X	X	4.4.2
3.4	Hygroscopicity	X		X	X	X	X	4.4.3
3.5	Acidity	X		X	X	X	X	4.4.4
3.6	Water-insoluble matter	X		X	X	X	X	4.4.5
3.7	Grit	X		X	X	X	X	4.4.6
3.8	Chloride	X		X	X	X	X	4.4.7
3.9	Ammonium compounds	X		X	X	X	X	4.4.8
3.10	Barium	X		X	X	X	X	4.4.9
3.10	Calcium	X		X	X	X	X	4.4.9
3.10	Magnesium	X		X	X	X	X	4.4.9
3.10	Sodium	X		X	X	X	X	4.4.9
3.10	Iron	X		X	X	X	X	4.4.9
3.10	Bismuth	X		X	X	X	X	4.4.9
3.10	Lead	X		X	X	X	X	4.4.9
3.10	Copper	X		X	X	X	X	4.4.9
3.10	Sulfur	X		X	X	X	X	4.4.9 or 4.4.10
3.11	Granulation	X		X	X	X	X	4.4.11
3.12	Workmanship			X		X	X	4.4.12

4.1 Classification of inspection. The verification requirements specified herein are classified as follows.

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

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4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Sampling procedures shall be conducted in accordance with 4.3.2. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table III fails to meet any of the requirements.

TABLE III. First article inspection.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Strontium nitrate assay	3.2	4.4.1
102	Moisture	3.3	4.4.2
103	Hygroscopicity	3.4	4.4.3
104	Acidity	3.5	4.4.4
105	Water-insoluble matter	3.6	4.4.5
106	Grit	3.7	4.4.6
107	Chloride	3.8	4.4.7
108	Ammonium compounds	3.9	4.4.8
109	Barium	3.10	4.4.9
110	Calcium	3.10	4.4.9
111	Magnesium	3.10	4.4.9
112	Sodium	3.10	4.4.9
113	Iron	3.10	4.4.9
114	Bismuth	3.10	4.4.9
115	Lead	3.10	4.4.9
116	Copper	3.10	4.4.9
117	Sulfur	3.10	4.4.9 or 4.4.10
118	Granulation	3.11	4.4.11
Minor			
201	Workmanship	3.12	4.4.12

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph “Formation and identification of lots or batches.” A lot shall consist of strontium nitrate produced consecutively on one shift, by one manufacturer, at the same plant, using the same equipment, in accordance with one specification and revision under one continuous set of operating conditions, with no change in raw materials, provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot. Each batch shall consist of that quantity of strontium nitrate that has been subjected to the same chemical and physical unit operations intended to make the final product homogeneous.

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4.3.2 Strontium nitrate sampling. Sampling for inspection shall be performed at either the supplier location where the strontium nitrate is packaged, or at the receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Batch sampling at the manufacturer. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 500 grams shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Strontium nitrate shall be sampled using Verification Level II of MIL-STD-1916, based up on the number of shipping containers (e.g., drums, boxes, etc.) in a lot. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. The samples from each shipping container shall be placed in a secondary container and thoroughly mixed to form a composite sample of not less than 500 g for each strontium nitrate lot received. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.3.3 and 4.4. If any sample fails to meet any test requirement the lot represented by the sample shall be rejected.

4.3.3 Classification of characteristics.

4.3.3.1 Strontium nitrate.				
Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Strontium nitrate assay	4.3.2	3.2	4.4.1
102	Moisture	4.3.2	3.3	4.4.2
103	Hygroscopicity	4.3.2	3.4	4.4.3
104	Acidity	4.3.2	3.5	4.4.4
105	Water-insoluble matter	4.3.2	3.6	4.4.5
106	Grit	4.3.2	3.7	4.4.6
107	Chloride	4.3.2	3.8	4.4.7
108	Ammonium compounds	4.3.2	3.9	4.4.8
109	Barium	4.3.2	3.10	4.4.9
110	Calcium	4.3.2	3.10	4.4.9
111	Magnesium	4.3.2	3.10	4.4.9
112	Sodium	4.3.2	3.10	4.4.9
113	Iron	4.3.2	3.10	4.4.9
114	Bismuth	4.3.2	3.10	4.4.9
115	Lead	4.3.2	3.10	4.4.9
116	Copper	4.3.2	3.10	4.4.9
117	Sulfur	4.3.2	3.10	4.4.9 or 4.4.10
118	Granulation	4.3.2	3.11	4.4.11
Minor				
201	Workmanship	4.3.2	3.12	4.4.12

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4.4 Test methods and procedures. Unless otherwise specified, all chemicals and reagents shall be reagent grade or better, and water shall be distilled, deionized, or equivalent, conforming to ASTM D1193 Type I or Type II. Blank determinations shall be run and corrections shall be applied when necessary.

4.4.1 Strontium nitrate assay.

4.4.1.1 Strontium nitrate assay. All weight measurements shall be made to the nearest 0.1 milligram (mg). Place 10.0 ± 0.05 g of tungstic oxide into a platinum crucible and ignite at 700º ± 50ºC (1292º ± 90ºF) in a furnace for 1 hour. Cool in a desiccator for 60 ± 5 minutes and weigh. Place the crucible back into the furnace for an additional half hour, cool in a desiccator 60 ± 15 minutes, and weigh again. Repeat the procedure until constant weight is reached as shown by readings of not greater than 0.2 mg difference. Allow the furnace to cool to approximately room temperature and adjust the controls to 690 ± 50 ºC (1274 ± 90ºF). Mix a 2 g portion of the sample (accurately weighed to 0.1 mg) with the ignited tungstic oxide, cover the crucible with a tight fitting cover, and weigh to the nearest 0.1 mg. Place the crucible in the furnace at approximately room temperature and heat to 690 ± 50 ºC (1274 ± 90ºF). Heat the crucible for 2 hours after the operating temperature has been reached, cool in a desiccator for 60 ± 5 minutes, and weigh. Reheat for an additional half hour, cool in a desiccator for 60 ± 5 minutes, and weigh again. If necessary, repeat the previous step until constant weight is reached as shown by readings of not greater than 0.2 mg difference. Calculate the percent strontium nitrate as follows:

$$\% \text{ strontium nitrate} = \frac{195.9 \times A}{B}$$

Where: A = weight of sample loss due to heating, g
B = weight of sample, g

4.4.1.2 Strontium nitrate assay (alternate method).). Weigh approximately 35 mg of strontium nitrate sample to the nearest 0.1 mg and dissolve in deionized water. Dilute this to one liter in a one liter volumetric flask. The sample should be analyzed in triplicate. After setting up the Lambda 5 uv-visible spectrophotometer and calibrating it to read out in absorbance according to the nitrate method, analyze each solution of strontium nitrate in turn. Record the readings obtained. Calculate the percent strontium nitrate as follows:

$$\% \text{ strontium nitrate} = \frac{\text{Abs of sample}}{\text{Abs of standard}} \times \frac{\text{mg } N_2 \text{ standard}}{\text{liter}} \times \frac{755.45}{C}$$

Where: Abs = absorbance measured at 220 nm
C = sample weight, mg

Note: 755.45 is calculated by dividing the molecular weight of strontium nitrate by the molecular weight of nitrogen (N_2) and taking that value times one hundred.

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4.4.1.3 Strontium nitrate assay (alternate method). Determine assay in accordance with ACS Reagent Chemicals, Assay Method.

4.4.2 Moisture. All weight measurements shall be made to the nearest 0.1 m
g. Transfer 6 grams of the sample to a pre-weighed aluminum weighing dish (approximately 6 cm in diameter and 2.5 cm deep) and record the gross weight. Dry at 130 ± 3˚C (266 ± 5˚F) for 3 hours, cool in a desiccator, and re-weigh. Calculate the percent moisture as follows:

$$\% \text{ moisture} = 100 \times \frac{E - F}{E - G}$$

Where: E = gross weight prior to drying, grams
F = gross weight after drying, grams
G = tare weight of aluminum dish, grams

Note: Save the dried samples for the hygroscopicity determination (see 4.4.3).

4.4.3 Hygroscopicity. All weight measurements shall be made to the nearest 0.1 mg. Place the dried samples from the moisture determination into a desiccator with 10 liter capacity, containing 1 liter of a sulfuric acid solution (33.1 ± 0.2% sulfuric acid by weight) for producing 70% relative humidity at 25ºC (77ºF). Allow the dishes to remain in the desiccator 48 hours and weigh. Place the dishes back into the desiccator for 24 hours and weigh again. Repeat for 24-hour intervals until constant weight is obtained as shown by weight changes of not greater than 0.2 mg over a 24-hour interval. Calculate the percent hygroscopicity as follows:

$$\% \text{ hygroscopicity} = 100 \times \frac{H - F}{E - G}$$

Where: E = original sample gross weight prior to drying, grams (see 4.4.2)
F = gross weight after drying and prior to humidification, grams (see 4.4.2)
G = tare weight of aluminum dish, grams (see 4.4.2)
H = gross weight after humidification, grams

4.4.4 Acidity (pH).

4.4.4.1 Equipment and reagents.

a. pH-meter with temperature compensation
b. pH combination electrode, double-junction, and stirrer
c. Water, deionized (DI) or distilled, conforming to ASTM D1193 Type I or II, freshly boiled or degasified
d. Solutions and reagents required for proper functioning of the pH electrode, per electrode operating manual
e. Standard pH buffer solutions, e.g. pH 4, 7, and 10, or another combination of three buffers solutions that bracket the 6-7 pH range.

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4.4.4.2 Procedure.

a. Prepare the pH electrode according to the manufacturer's operating manual.

b. Calibrate the pH-meter in accordance with manufacturer's manual using the standard buffer solutions; at least three (3) different pH standard buffer solutions (see 4.4.4.1.e) shall be used. The electrode shall be rinsed with water between measurements.

c. The pH of the calibration standard closest to the 6-7 pH range (e.g. pH 7 standard buffer) shall be measured before and after making sample measurements. The standard buffer shall not differ from its certified value by more than 0.05 pH units. If the standard buffer does vary by greater than 0.05 pH units it shall be replaced with new buffer and the electrode recalibrated (see also 6.6).

d. Weigh 5.0 ± 0.1 g of the strontium nitrate sample, add 75 mL of water (see 4.4.4.1.c) and stir until sample is dissolved. Dilute to 100 mL with additional water.

e. Rinse the electrode with an aliquot of sample solution, then place electrode in a fresh aliquot of solution and stir for 30 seconds. Allow the solution to come to rest and immediately record the pH.

f. Continue alternately stirring, resting, and recording the pH at 30 second intervals until two consecutive measurements differ by less than 0.1 pH units (around two minutes depending on electrode and sample).

4.4.5 Water-insoluble matter. Weigh 50 g of the sample into a 400 mL beaker, add about 300 mL of water and stir to dissolve. Heat to boiling and boil for 5 minutes with occasional stirring. Filter through a tared sintered glass crucible of medium porosity and wash with hot water. Dry the crucible at 105º ± 3 ºC (221±5ºF) for 1 hour. Cool in a desiccator and weigh. Calculate the percent water-insoluble matter as follows:

$$\text{Percent water-insoluble matter} = \frac{100 \times J}{K}$$

Where: J = weight of residue, g
K = weight of sample, g

4.4.6 Grit. Weigh 50 g of the sample into a 400 mL beaker, add about 300 mL of water and stir to dissolve. Heat to boiling and boil for 5 minutes with occasional stirring. Filter through a Whatman No. 40 filter paper, transfer and wash with hot water. Save the filtrate for a determination of chloride (see 4.4.7.1). Wash the residue from the filter paper into a 100-mL beaker with a stream of water from a wash bottle. Add 10 mL of nitric acid to the 100-mL beaker and evaporate to dryness on a steam bath. Add about 10 mL of water and break up any residue with a policeman. Transfer to a U.S. Alternative No. 170 sieve with water. If any solid material is retained by the sieve, wash well with water, alcohol, and finally ethyl ether. Transfer the material to a tared weighing dish and weigh. Calculate the percent grit as follows:

$$\text{Percent grit} = \frac{100 \times L}{M}$$

Where: L = weight of residue, g

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M = weight of sample, g

4.4.7 Chloride.

4.4.7.1 Chloride content by precipitation. Add 5 mL of concentrated 70% nitric acid to the filtrate from the grit determination (see 4.4.6). Heat to boiling, add 5 mL of 5% silver nitrate solution, and boil for 5 minutes. Allow to stand in a dark place overnight at room temperature. Filter through a tared sintered glass crucible of fine porosity and wash with water. Dry at 150º ± 5 ºC (302º ± 9ºF) for 1 hour, cool in a desiccator and weigh. Calculate the percent chloride as follows:

$$\text{Percent chloride (as Cl)} = \frac{24.7 \times N}{M}$$

Where: N = weight of residue, g
M = weight of sample, g (see 4.4.6)

4.4.7.2 Chloride content by turbidity (alternate). Determine chloride content in accordance with ACS Reagent Chemicals, Chloride Method for strontium nitrate.

4.4.8 Ammonium compounds.

a. Weigh 10 g of the sample into a 500 mL Kjeldahl flask and dissolve in 100 mL of deionized water.
b. Add 10 mL of 20% sodium hydroxide solution and immediately connect to the Kjeldahl distillation apparatus, the outlet tube of which dips below 50 mL of 3% boric acid solution contained in a 250 mL Erlenmeyer flask (extend the outlet with PTFE tubing if necessary). Ensure all connections are tight.
c. Distill about 75 mL, wash down the outlet tube, add 3 drops of methyl purple indicator, and seal the flask.
d. Carry a blank, i.e. 100 mL water, through steps a-c.
e. View the color of collected distillates against white background. If both sample and blank are purple of similar saturation then ammonium compounds content is nil. If the sample or blank is very light violet, nearly colorless, or light green, ammonium compounds are present; proceed with remaining steps.
NOTE: Clean the Kjeldahl flask from precipitated strontium hydroxide with dilute hydrochloric acid, then rinse with deionized water until pH is 5-8 to litmus.
f. Titrate first the blank distillate and then the sample distillate with standard 0.05 N sulfuric acid solution until the color changes from green to light purple. Use the blank as endpoint color reference when titrating the sample.
g. Calculate the percent ammonium compounds as follows:

$$\text{Percent ammonium compounds (as } NH_3) = \frac{1.70 \times (S - T) \times U}{V}$$

Where: S = sulfuric acid for titration of sample, mL
T = sulfuric acid for titration of blank, mL

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U = normality of sulfuric acid
V = weight of sample, g

4.4.9 Barium, calcium, magnesium, sodium, iron, bismuth, lead, copper, and sulfur by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES) method.

4.4.9.1 Reagents. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts; these materials shall be Reagent grade or better. The maximum recommended shelf life is one year from date of initial preparation when kept tightly sealed at 5-25°C. Common commercially available stock concentrations are as follows:

a. Certified Ba stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately dissolve 1.903 ± 0.005 g of barium nitrate in nitric acid 10 % v/v, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark.

b. Certified Bi stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately weigh 1.000 ± 0.005 g of bismuth metal, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

c. Certified Ca stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately dissolve 2.497 ± 0.005 g of calcium carbonate in 10% v/v nitric acid, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

d. Certified Cu stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately weigh 1.000 ± 0.005 g copper metal, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

e. Certified Fe stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately weigh 1.000 ± 0.005 g of elemental iron, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

f. Certified Mg stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately weigh 1.000 ± 0.005 g of elemental magnesium, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

g. Certified Na stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately dissolve 2.305 ± 0.005 g of sodium carbonate in nitric acid 10 % v/v, cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark.

h. Certified Pb stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO_3 or HCl; alternately weigh 1.000 ± 0.005g of elemental lead, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

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i. Certified S stock solution, 1000 ± 5 mg/L in water (where sulfur is in the form of sulfate, and no metals are present); alternately dissolve 4.121 ± 0.005 g of ammonium sulfate in water, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

j. Certified Y stock solution, 1000 ± 5 mg/L in 1-10% (v/v) HNO3 or HCl; alternately weigh 1.000 ± 0.005 g of elemental yttrium, add approx. 50 mL water, then slowly add 200 mL of 50 % v/v nitric acid until fully dissolved (heat if necessary), cool, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

Additional necessary reagents are:

k. Nitric acid (HNO_3), concentrated, 67-70%, trace metal grade.

l. Nitric acid (HNO_3), 5%, made by diluting 50 ± 2 mL of concentrated nitric acid to 1 L with water.

m. Nitric acid (HNO_3), 10%, made by diluting 100 ± 2 mL of concentrated nitric acid to 1 L with water.

n. Nitric acid (HNO_3), 50%, made by diluting 500 ± 2 mL of concentrated nitric acid to 1 L with water.

o. Water, de-ionized or distilled, conforming to ASTM D1193 Type I or II shall be used throughout.

p. Bromine, reagent-grade.

4.4.9.2 Apparatus and supplies.

a. Inductively coupled, argon-plasma, optical emission spectrometer, dual view, e.g.,ThermoFisher iCAP6500 ICP system with iTeva version 2.5 software or equivalent

b. Compressed argon, as per ICP-OES manufacturer recommendations

c. Class A volumetric flasks and pipettes (see 6.7)

d. 0.1 - 1 mL and 1 - 5 mL adjustable volume pipettes (calibration shall have been verified not greater than one month prior)

e. Analytical balance with 200 g capacity and 0.1 mg resolution.

4.4.9.3 Preparation of samples and standards solutions, and reagent blanks.

Note: The calibration and analysis are performed using the method of standard additions (MSA). Yttrium is used as an internal standard to control unwanted variation.

4.4.9.3.1 For Grade A (all analytes) and Grade B (all analytes except barium).

a. Weigh 3.000 ± 0.002 g of sample into each of seven (7) 250 mL conical flasks.

b. Add the amounts of stock solutions indicated in Table IV to five of the conical flasks to serve as MSA calibration standards (calibration spikes 1-3) and independent check samples 1 & 2.

c. Add 50 mL of water, 5 mL of concentrated nitric acid, and 0.5 mL bromine. Mix well and heat the solutions over a hotplate (250-280˚C surface temperature) for about 30 minutes. The solutions should become colorless without any black droplets of bromine.

d. Allow the solutions to cool. Add the yttrium stock solution per Table IV and quantitatively transfer from the conical flasks to 100 mL volumetric flasks. Make up to volume with water and mix well.

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e. A reagent blank shall be carried through the entire procedure and its reading subtracted from all samples, either automatically by ICP software or manually.

Table IV. Volumes (mL) of each 1000 mg/L stock indicated for sample preparation of Grade A material (or Grade B material for all analytes except barium). 1/

Sample identifier	Weight, g	Ba	Bi	Ca	Cu	Fe	Mg	Na	Pb	S	Y 2/
Reagent blank	0	0	0	0	0	0	0	0	0	0	0.100
Calibration spike 0 (replicate 1)	3.000 ± 0.002	0	0	0	0	0	0	0	0	0	0.100
Calibration spike 0 (replicate 2)	3.000 ± 0.002	0	0	0	0	0	0	0	0	0	0.100
Calibration spike 1	3.000 ± 0.002	0.50	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.30	0.100
Calibration spike 2	3.000 ± 0.002	2.00	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.60	0.100
Calibration spike 3	3.000 ± 0.002	3.50	0.70	0.70	0.70	0.70	0.70	0.70	0.70	1.00	0.100
Check sample 1	3.000 ± 0.002	1.50	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.40	0.100
Check sample 2	3.000 ± 0.002	1.50	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.40	0.100

Notes:
1/ Actual concentrations of commercially obtained stock solutions or of those prepared in-house shall be used. Added concentrations in Table V shall be adjusted proportionally.
2/ Yttrium stock solution is added after solution has cooled and just before transferring to 100 mL flask.

Table V. Nominal added concentrations (mg/L) when samples prepared per Table IV.

Sample identifier	Ba	Bi	Ca	Cu	Fe	Mg	Na	Pb	S	Y
Reagent blank	0	0	0	0	0	0	0	0	0	1
Calibration spike 0 (replicate 1)	0	0	0	0	0	0	0	0	0	1
Calibration spike 0 (replicate 2)	0	0	0	0	0	0	0	0	0	1
Calibration spike 1	5	1	1	1	1	1	1	1	3	1
Calibration spike 2	20	4	4	4	4	4	4	4	6	1
Calibration spike 3	35	7	7	7	7	7	7	7	10	1
Check sample 1	15	1.5	1.5	1.5	1.5	1.5	1.5	1.5	4	1
Check sample 2	15	1.5	1.5	1.5	1.5	1.5	1.5	1.5	4	1

4.4.9.3.2 For Grade B (barium).

a. Weigh 0.100 ± 0.002 g of sample into each of seven (7) 250 mL conical flasks.
b. Add the amounts of stock solutions indicated in Table VI to five of the conical flasks to serve as MSA calibration standards and independent check solutions..
c. Add 50 mL of water and 5 mL of concentrated nitric acid. Mix well to dissolve.

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d. Add the yttrium stock solution per Table VI and quantitatively transfer from the conical flasks to 100 mL volumetric flasks. Make up to volume with water and mix well.

e. A reagent blank shall be carried through the entire procedure and its reading subtracted from all samples, either automatically by ICP software or manually.

Table VI. Volumes (mL) of each 1000 mg/L stock indicated for sample preparation of Grade B material for barium determination. 1/

Sample identifier	Weight, g	Ba	Y 2/
Reagent blank	0	0	0.100
Calibration spike 0 (replicate 1)	0.100±0.002	0	0.100
Calibration spike 0 (replicate 2)	0.100±0.002	0	0.100
Calibration spike 1	0.100±0.002	0.50	0.100
Calibration spike 2	0.100±0.002	2.00	0.100
Calibration spike 3	0.100±0.002	3.50	0.100
Check sample 1	0.100±0.002	1.50	0.100
Check sample 2	0.100±0.002	1.50	0.100

Notes:
1/ Actual concentrations of commercially obtained stock solutions or of those prepared in-house shall be used. Added concentrations in Table VII shall be adjusted proportionally.
2/ Yttrium stock solution is added after solution has cooled and just before transferring to 100 mL flask.

Table VII. Nominal added concentrations (mg/L) for sample preparation of Grade B material for barium determination.

Sample identifier	Ba	Y
Reagent blank	0	1
Calibration spike 0 (replicate 1)	0	1
Calibration spike 0 (replicate 2)	0	1
Calibration spike 1	5	1
Calibration spike 2	20	1
Calibration spike 3	35	1
Check sample 1	15	1
Check sample 2	15	1

4.4.9.4 Instrument preparation. Prepare the instrument according to manufacturer instructions. If the recommended instrument and software (4.4.9.2.a) are not used then additional method development may be required. The wavelengths used for analyzing each element are as follows in Table VIII:

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TABLE VIII. Recommended wavelengths and view modes for ICP-OES element analysis.

Element	Wavelength, nanometers (nm)	View	Referencing elements
Ba	455.4 or 493.4 1/	radial	
Ca	393.3	radial	
Bi	223.0	axial	
Cu	224.7	axial	
Fe	238.2	axial	
Mg	280.2	axial	
Na	589.5	radial	
Pb	220.3	axial	
S	182.0	axial	
Y (internal standard)	224.3	axial	Bi, Cu, Pb, S
Y (internal standard)	371.0	axial	Fe, Mg
Y (internal standard)	371.0	radial	Ba, Ca, Na
Notes: 1/ Either 455.4 nm or 493.4 nm may be used for analysis of barium			

4.4.9.5 Analysis. Due to the high solids content and low analyte concentrations, the sample introduction system shall be rinsed with 5 % nitric acid before the analysis and between samples.

a. Perform calibration in standard additions mode using the following sampling order: reagent blank; calibration spike 0 (replicate 1); calibration spike 1; calibration spike 2; calibration spike 3. The generated calibration shall be linear with $R^2 >= 0.999$; if not then repeat 4.4.9.3. (The software will calculate and report analyte concentrations in calibration spike 0 (rep 1). Alternately, or as a check, the concentration may be calculated outside the software, using commonly accepted evaluation of MSA.)

b. Repeat the sampling sequence for calibration spike 0 (replicate 2) and obtain a second set of analyte concentrations.

c. Results of the two replicates of calibration spike 0 shall differ by not greater than 1.5 mg/L for barium in Grade A material or 0.50 mg/L barium in Grade B material, and not greater than 0.30 mg/L for sulfur and 0.15 mg/L for the other analytes in either grade.

d. Validate the results by analyzing check samples 1 and 2 using the internal calibration (i.e. response relative to yttrium versus added concentration) derived from either 4.4.9.5.a or b. The reagent blank shall be run, and its readings subtracted from those of the check samples prior to quantitation.

e. If derived by the software, ensure that check sample concentration represents only the spike (added) concentration rather than the total analyte concentration. Check sample results shall be within 10 % relative deviation of the expected values (per Table IV or VII) for all elements.

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If all above conditions are met, calculate the average concentration of each analyte in calibration spike 0 based on replicates 1 and 2.

f. Calculate the weight percent of each elemental impurity as follows:

$$\text{Weight \%} = \frac{0.01 \times A}{B}$$

Where: A = average (mean) analyte concentration in calibration spike 0, mg/L
B = sample mass, g

4.4.10 Sulfur (total) (alternate). Weigh 20 g of the sample, into a 400 mL beaker, add about 200 ml. of water, and stir to dissolve. Add 2 mL of hydrochloric acid and 10 mL of bromine water. Heat to boiling and boil 10 minutes. Filter through a Whatman No. 40 filter paper and wash with water. Add 10 mL of 10% barium chloride solution, heat to boiling, and boil 5 minutes. Allow to stand overnight. Filter through a Whatman No. 42 filter paper, transfer, and thoroughly wash with water. Place the filter paper and precipitate into a tared platinum crucible. Char and burn off the filter paper at dull red heat and then ignite at red heat for 30 minutes. Cool in a desiccator and weigh. Calculate the percent sulfur as follows:

$$\text{Percent sulfur (total)} = \frac{13.7 \times C}{D}$$

Where: C = weight of precipitate, g
D = weight of sample, g

4.4.11 Granulation. Dry the material to be sieved by spreading in a thin layer on a large watch glass and heating in an oven at 57º ± 3 ºC (135º ± 5ºF) for 1 hour. For Grade A, 100.0 ± 2.0 g of the dried material shall be transferred to a U.S. Alternative No. 100 sieve fitted with a bottom pan. For Grade B, 100.0 ± 2.0 of the dried material shall be transferred to a U.S. Alternative No. 50 U.S. standard sieve nested over a U.S. Alternative No. 140 sieve fitted with a bottom pan. The sieve or sieve assembly shall be provided with a cover. The sieves shall conform to ASTM E11. Shake for 5 minutes by means of a mechanical shaker geared to produce 268 to 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the retained portions (to the nearest 0.01 g) on each sieve and calculate to percent passed.

4.4.12 Workmanship. Take approximately 300 grams of the material and spread the composition out on a clean white surface. The sample shall be examined by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from lumps, visible impurities, or foreign matter. Failure to meet the workmanship inspection or during following production processes shall result in batch/lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory

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Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Strontium nitrate Grade A and Grade B are for use as follows:

Grade A – for use in pyrotechnic compositions, such as flares.
Grade B – for use in tracer compositions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Grade of material required (see 1.2).
c. Packaging requirements (see 5.1).
d. Requirements for First article samples (see 4.2).
e. Provision for submission of first article samples.
f. Requirements for submission of acceptance inspection equipment.
g. Requirements for ammunition lot numbering.

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to the Technical Agency (see 6.5).

6.4 Equivalent test method approval. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision, and drawings of any special equipment required.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM,Armaments Center, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.6 Acidity calibration. As an independent verification of pH system performance, using the above procedure 4.4.4, measure the pH of a 5% solution of ACS-grade strontium nitrate, whose pH is known from a certificate of analysis. The measured pH should not differ from its certified value by more than 0.20 pH units.

6.7 Class A glassware. Class A glassware should meet the requirements of ASTM E694 "Standard Specification for Laboratory Glass Volumetric Apparatus" and ASTM E542 "Standard Practice for Calibration of Laboratory Volumetric Apparatus."

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6.8 Subject term (key word) listing.

Pyrotechnic composition
Tracer composition

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – AS

Preparing Activity:
Army – AR
(Project 6810-2023-008)

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